Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech (ME) Examination

NOV 2018

ME417 Internal Combustion Engines

Date: 30.11.2018, Friday **Time: 01.30 p.m. To 04.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 **Answer the question below.** **[05]**

a. Which pollutant is emitted as a evaporative loss from engine

1. HC 2. CO 3. NOx 4. PM

b. Combustion in spark ignition engine is ______.

1. Homogeneous 2. Heterogeneous 3. laminar 4. None of above

c. The A/F ratio for idling speed of an automobile petrol engine is closer to

1. 21:1 2. 17:1 3. 15:1 4. 10:1

d. Rating of SI engine is done by

1. Cetane number 2. Octane number 3. Both of above 4. None of above

e. By increasing surface to volume ratio HC emission

1. Increase 2. Decrease 3. Remains same 4. Any of above

Q-2.(a) What is meant by following terms in I.C. engine fuels. **[05]**

(i) Viscosity (ii) Volatility (iii) Flash point (iv) Fire point (v) Cloud point

Q-2.(b) Show that if $c_p = a + kT$ and $c_v = b + kT$ the adiabatic expansion of the gas is represented by **[06]**

$$\boldsymbol{p^b v^a e^{kT} = constant}$$

OR

(b) How the variation of specific heats takes place with temperature? What is the physical explanation for this variation? **[06]**

(c) Discuss effects of following factors on delay period in CI engines **[06]**

(i) Compression ratio (ii) Quality of Fuel (iii) Intake pressure

OR

(c) What are the different factors affecting combustion chamber design in S.I. engines? **[06]**

Q-3.(a) Explain phenomenon of Pre-ignition in SI engine. Does it cause knocking in the engines? Justify. **[07]**

OR

(a) What is knock rating for C.I.engine fuels? Explain various methods for controlling knock. **[07]**

(b) Explain eddy current dynamometer along with construction, working, merits and demerits. **[06]**

OR

(b) Explain sources of HC emissions from engines with its effects and method of measurement of HC. **[06]**

SECTION – II

Q - 4 **Answer the question below.** **[05]**

a. For Provision of cold starting and warming up, Solex carburetor has a provision of

1. Choke 2. Venturi 3. Bi-starter 4. Emulsion tube

b. Which one is not a function of fuel injection system

1. Fuel metering 2. Fuel pressurization 3. Fuel atomization 4. Fuel Ignition

c. A diesel engine has compression ratio from

1. 20 to 30 2. 16 to 20 3. 10 to 15 4. 6 to 10

d. Which of the following is not a mechanical type tachometer

1. slipping clutch tachometer 2. Centrifugal force tachometer
3. Drag cup tachometer 4. Tachoscope

e. In CRDI engine, HP pump supplies fuel to

1. LP pump 2. Nozzle 3. Common header 4. Inlet valve

Q-5.(a) Explain S U carburetor in detail with neat sketch **[07]**

OR

(a) Derive an expression for fuel/ air ratio for a carbureted engine taking compressibility in account. **[07]**

(b) Discuss the basic requirements of a spark ignition system. Explain the use of distributor and ballast resistor in battery ignition system. **[07]**

OR

(b) Describe the principle of helix bypass pump and also draw sketches for different types of plunger helix in use. **[07]**

(c) Compare air and solid injection system for CI engines. Discuss the requirements of an ideal injection system. **[06]**

Q-6.(a) What is supercharging? How it is achieved? Describe the Buchi system of supercharging **[06]**

OR

(a) What is meant by pulse turbocharging? What are its advantages and disadvantages? **[06]**

(b) Explain working principle of piston less engine. **[04]**

***